

Note: There are **three** sections (A, B and C) and candidate has to attempt questions from all sections. Marks are indicated against each section.

Section-A

1. Attempt **all** parts of the following : 5×3=15

(a) What is Machine Learning ? Describe its types.

(b) If $A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & -1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$,
find $2A - 3B$.

(c) If $X \sim \text{Bin}(n, P)$, show that :

(i) $E \left(\frac{X}{n} - P \right)^2 = \frac{Pq}{n}.$

(ii) $\text{Cov} \left(\frac{X}{n}, \frac{n-X}{n} \right) = \frac{-Pq}{n}.$

(d) Define the following terms :

(i) Quartile deviation

(ii) Mean deviation

(iii) Standard deviation

- (e) Show that the set $W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}$ is a subspace of $\mathbb{R}^3$.

Section-B

Note: Answer **all** questions of the following :

4×5=20

2. (a) Solve the equation :

$$\log_{10}(3x+2) - 2\log_{10}x = 1 - \log_{10}(5x-3).$$

Or

- (b) If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = 0$. Hence find A^{-1} .

3. (a) Show that :

$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3.$$

Or

- (b) Find local maximum and local minimum values of the function.

$$f(x) = \sin x + \cos x, \quad 0 < x < \frac{\pi}{2}.$$

4. (a) If X is a Poisson variate such that

$$P(X=2) = 9P(X=4) + 90P(X=6)$$

find :

- (i) Mean of X
- (ii) β_1 , the coefficient of skewness

Or

- (b) Mention the chief characteristics of normal distribution and normal probability curve.

5. (a) A r.v. X has the pdf :

$$f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases} \quad \text{Find } P\left(X < \frac{3}{4} \mid X > \frac{1}{2}\right).$$

Or

- (b) Calculate Arithmetic Mean (AM) and Standard Deviation (S.D) from the following data :

X	2	4	6	8	10
f	1	4	6	4	1

Section-C

Note: Answer any **two** questions of the following :

$2 \times 20 = 40$

6. (a) Find the LU decomposition of the following matrix :

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$$

- (b) Find the gradient and Hessian matrix of the function.

$$f(x, y, z) = x^2 + y^2 + yz \sin(x).$$

7. (a) Solve the following system of equations by matrix method.

$$3x_1 - 2x_2 + 3x_3 = 8$$

$$2x_1 + x_2 - x_3 = 1$$

$$4x_1 - 3x_2 + 2x_3 = 4.$$

- (b) Find the quotient and remainder when the polynomial $P(x) = 4x^3 + 3x^2 - 3x + 1$ is divided by $D(x) = x - 2$.
8. (a) Describe the maximum likelihood (M.L) method of estimation and discuss five of its optimal properties. Also give an example.
- (b) If $x \geq 1$ is the critical region for testing $H_0 : \theta = 2$ against the alternative $H_1 : \theta = 1$, on the basis of a single observation from the population.
- $$f(x, \theta) = \theta e^{-\theta x}; x > 0, \theta > 0.$$
- Obtain the value of probability of Type-I and Type-II errors and also power function.
9. (a) Define the “Distribution function” of a random variable and state its important properties.
- (b) Let a bivariate r.v. (X, Y) has joint pdf :
- $$f(x, y) = \begin{cases} 8xy, & 0 < x < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$
- Find –
- Marginal density $f_x(x)$ and $f_y(y)$.
 - $E(Y | X = \frac{1}{2})$
 - $V(Y | X = \frac{1}{2})$

